Pure Mathematics U2 2026 P2 Solutions

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MODULE 1

Question 1

Question 1: (a)

Express the complex number $\dfrac{5-3i}{4+2i}$ in the form $x+yi$ where x and y are constants. **[4 marks]**

Solution 1: (a)

Let: $z = \dfrac{5-3i}{4+2i}$

$z = \dfrac{5-3i}{4+2i} \times \dfrac{4-2i}{4-2i}$ $\quad \times\ \overline{z}$

$z = \dfrac{20-10i-12i+6i^2}{16-8i+8i-4i^2}$ $\quad$ Re: $i^2 = -1$

$z = \dfrac{14-22i}{16+4}$

$z = \dfrac{14-22i}{20}$

$z = \dfrac{7}{10} - \dfrac{11}{10}i$

Question 1: (b)(i)

Let: $z = -1 + \sqrt{3}i$

Calculate the EXACT value of $|z|$. **[2 marks]**

Solution 1: (b)(i)

$|z| = \sqrt{(-1)^2 + (\sqrt{3})^2}$

$|z| = \sqrt{4}$

$|z| = 2$ $\qquad |z| \in \mathbb{R}^+$

Question 1: (b) (i) (b)

$\arg(z)$ **[2 marks]**

Solution 1: (b) (i) (b)

$\arg(z) = \pi - \alpha$ $\qquad$ As z lies within the 2^{nd} Quadrant

Where: $\alpha = \arctan\left(\frac{\sqrt{3}}{1}\right)$

$\alpha = \frac{\pi}{3}$

$\arg(z) = \pi - \frac{\pi}{3}$

$\arg(z) = \frac{2\pi}{3}$

Question 1: (b) (ii)

Hence, use de Moivre's theorem to determine the value of z^{12}. **[4 marks]**

Solution 1: (b) (ii)

Re: $z = -1 + \sqrt{3}i$,$|z| = 2$, $\arg(z) = \dfrac{2\pi}{3}$

$$z = 2\left(\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3}\right) \qquad \text{is of Polar Form}$$

$$z^{12} = \left[2\left(\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3}\right)\right]^{12}$$

$$z^{12} = 2^{12}\left(\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3}\right)^{12}$$

$$z^{12} = 4096\left(\cos\frac{24\pi}{3} + i\sin\frac{24\pi}{3}\right) \qquad \text{By de Moivre's Theorem}$$

$$z^{12} = 4096(1+0)$$

$$z^{12} = 4096$$

Question 1: (c)

The equation of a curve is $y^3 + 3x^2y + x = 6$. Determine an expression for $\dfrac{dy}{dx}$ in terms of x and y. **[6 marks]**

Solution 1: (c)

$$y^3 + 3x^2y + x = 6$$

$$3y^2\frac{\mathrm{d}y}{\mathrm{d}x} + 3\left[2xy + x^2\frac{\mathrm{d}y}{\mathrm{d}x}\right] + 1 = 0$$

$$3y^2\frac{\mathrm{d}y}{\mathrm{d}x} + 6xy + 3x^2\frac{\mathrm{d}y}{\mathrm{d}x} + 1 = 0$$

$$\frac{\mathrm{d}y}{\mathrm{d}x}\left[3y^2 + 3x^2\right] + 6xy + 1 = 0$$

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{-(6xy+1)}{3(y^2+x^2)}$$

Question 1: (d)

Show that $y = -x + \frac{\pi}{2}$ is the tangent to the curve $y = \ln\sqrt{1+\sin 2x}$ at the point where $x = \frac{\pi}{2}$. **[7 marks]**

Solution 1: (d)

R.T.P: $y = -x + \frac{\pi}{2}$

Proof:

$$y = \frac{1}{2} ln\left(1+\sin 2x\right)$$

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{1}{2}\left[\frac{2\cos 2x}{1+\sin 2x}\right]$$

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{1}{2}\left[\frac{2\cos 2x}{1+\sin 2x}\right]$$

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{\cos 2x}{1+\sin 2x}$$

When $x = \frac{\pi}{2}$:

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{\cos \pi}{1+\sin \pi}$$

$$\frac{\mathrm{d}y}{\mathrm{d}x} = -1$$

$$y = \frac{1}{2} ln\left(1+\sin \pi\right)$$

$$y = 0$$

$$\implies \left(\frac{\pi}{2}, 0\right)$$

$$y - 0 = -1\left(x - \frac{\pi}{2}\right)$$

$$y = -x + \frac{\pi}{2}$$

Q.E.D.

Re: $y - y_1 = \frac{\mathrm{d}y}{\mathrm{d}x}\left(x - x_1\right)$

Question 2

Question 2: (a)(i)

Apply the trapezium rule with THREE equal intervals to estimate the value of $\displaystyle\int_0^{\pi/3} \tan x \, dx$ **[4 marks]**

Solution 2: (a)(i)

Re: $\int_a^b f(x)\,dx \approx \frac{h}{2}\left[y_0 + y_n + 2(y_1 + y_2 + \cdots + y_{n-1})\right]$

$$h = \frac{\frac{\pi}{3} - 0}{3} = \frac{\pi}{9}$$

x	0	$\frac{\pi}{9}$	$\frac{2\pi}{9}$	$\frac{\pi}{3}$
$f(x)$	0	0.3640	0.8391	$\sqrt{3}$

$$\int_0^{\pi/3} \tan x \, dx \approx \frac{1}{2} \times \frac{\pi}{9}\left[0 + \sqrt{3} + 2(0.3640 + 0.8391)\right]$$

$$\int_0^{\pi/3} \tan x \, dx \approx 0.722 \text{ (3 s.f.)}$$

Question 2: (a)(ii)

Using an appropriate trigonometric identity, determine the EXACT value of $\displaystyle\int_0^{\pi/3} \tan x \, dx$ **[4 marks]**

Solution 2: (a)(ii)

$$-\int_0^{\pi/3} \frac{-\sin x}{\cos x}\,dx \qquad \text{Re: } \tan x = \frac{\sin x}{\cos x}$$

$$= \left[\ \ln|\sec x|\ \right]_0^{\pi/3} \qquad \text{Re: } -\ln\cos x = \ln\sec x$$

$$= \left[\ln\sec\frac{\pi}{3}\right] - [\ln\sec 0]$$

$$= \ln 2 - \ln 1 \qquad \frac{1}{\cos\frac{\pi}{3}} = 2 \qquad \frac{1}{\cos 0} = 1$$

$$= \ln 2 \text{ (exact form)}$$

Question 2: (b)

Use the substitution $u = \cos^{-1}\left(\frac{1}{2}x\right)$ to show that $\displaystyle\int_0^1 \frac{\cos^{-1}\left(\frac{1}{2}x\right)}{\sqrt{4-x^2}}\,\mathrm{d}x = \frac{5\pi^2}{72}$. **[7 marks]**

Solution 2: (b)

R.T.P: $\displaystyle\int_0^1 \frac{\cos^{-1}\left(\frac{1}{2}x\right)}{\sqrt{4-x^2}}\,\mathrm{d}x = \frac{5\pi^2}{72}$

Proof:

$u = \cos^{-1}\left(\frac{1}{2}x\right)$ $\qquad u = \cos^{-1}\left(\frac{1}{2}\times 0\right) = \frac{\pi}{2}$ $\qquad u = \cos^{-1}\left(\frac{1}{2}\times 1\right) = \frac{\pi}{3}$

$$\frac{\mathrm{d}u}{\mathrm{d}x} = -\frac{\frac{1}{2}}{\sqrt{1-\left(\frac{x}{2}\right)^2}}$$

$$\frac{\mathrm{d}u}{\mathrm{d}x} = -\frac{2}{2\sqrt{4\left(1-\left(\frac{x}{2}\right)^2\right)}} \qquad -\frac{1}{2\sqrt{1-\left(\frac{x}{2}\right)^2}}\times\frac{\sqrt{4}}{\sqrt{4}}$$

$$\frac{\mathrm{d}u}{\mathrm{d}x} = -\frac{1}{\sqrt{4-x^2}}$$

$$-\,\mathrm{d}u = \frac{1}{\sqrt{4-x^2}}\,\mathrm{d}x \qquad \text{Substitute into } \int$$

$$-\int_{\frac{\pi}{2}}^{\frac{\pi}{3}} u\,\mathrm{d}u$$

$$= -\left[\frac{u^2}{2}\right]_{\frac{\pi}{2}}^{\frac{\pi}{3}}$$

$$= -\left[\frac{\left(\frac{\pi}{3}\right)^2}{2} - \frac{\left(\frac{\pi}{2}\right)^2}{2}\right]$$

$$= \frac{\pi^2}{8} - \frac{\pi^2}{18}$$

$$= \frac{5\pi^2}{72}$$

Q.E.D.

Question 2: (c)(i)

It is given that $I_n = \displaystyle\int_0^{\pi} \sin^n \theta \, \mathrm{d}\theta$ for integers $n \geqslant 0$.

Show that $I_n = \dfrac{n-1}{n} I_{n-2}$ **[6 marks]**

Solution 2: (c)(i)

$$I_n = \int_0^{\pi} \sin^{n-1} \theta \cdot \sin \theta \, \mathrm{d}\theta$$

Let:

$u = \sin^{n-1} \theta$ $\qquad$ $\mathrm{d}v = \sin \theta \, \mathrm{d}\theta$

$\dfrac{\mathrm{d}u}{\mathrm{d}\theta} = (n-1) \cos \theta \sin^{n-2} \theta$ $\qquad$ $v = -\cos \theta$

$$I_n = -\cos \theta \sin^{n-1} \theta \,\Big|_0^{\pi} - (n-1) \int_0^{\pi} -\cos^2 \theta \sin^{n-2} \theta \, \mathrm{d}\theta$$

$$I_n = 0 + (n-1) \int_0^{\pi} \left(1 - \sin^2 \theta\right) \sin^{n-2} \theta \, \mathrm{d}\theta \qquad \sin^{n-1}(0) = 0 \qquad \sin^{n-1}(\pi) = 0$$

$$I_n = (n-1) \int_0^{\pi} \sin^{n-2} \theta - \sin^n \theta \, \mathrm{d}\theta$$

$$I_n = (n-1)\left(I_{n-2} - I_n\right)$$

$$I_n + (n-1)\left(I_n\right) = (n-1) I_{n-2}$$

$$I_n (1 + n - 1) = (n-1) I_{n-2}$$

$$I_n = \frac{n-1}{n} I_{n-2} \qquad n \geqslant 2$$

Q.E.D.

Question 2: (ii)(a)

Verify that $I_1 = 2$. **[2 marks]**

Solution 2: (ii)(a)

$$I_n = \int_0^{\pi} \sin^n \theta \, \mathrm{d}\theta$$

When $n = 1$:

$$I_1 = \int_0^{\pi} \sin \theta \, \mathrm{d}\theta$$

$$I_1 = [-\cos \theta]_0^{\pi}$$

$$I_1 = [-\cos \pi] - [-\cos 0]$$

$$I_1 = -(-1) + 1$$

$$I_1 = 2$$

Q.E.D.

Question 2: (ii)(b)

Hence, determine the value of I_7 **[2 marks]**

Solution 2: (ii)(b)

$$I_7 = \frac{6}{7} I_5 \implies I_5 = \frac{4}{5} I_3 \implies I_3 = \frac{2}{3} I_1$$

Re: $I_1 = 2$

$$I_3 = \frac{2}{3} \times 2 = \frac{4}{3}$$

$$I_5 = \frac{4}{5} \times \left(\frac{4}{3}\right) = \frac{16}{15}$$

$$I_7 = \frac{6}{7} \left(\frac{16}{15}\right)$$

$$I_7 = \frac{32}{35}$$

Module 2

Question 3

Question 3: (a)(i)

Consider the sequence $\{a_n\}$ where

$$a_n = \frac{1}{3^{n-1}}$$

for positive integers n.

state the values of the first four terms of the sequence, a_1, a_2, a_3 and a_4 **[2 marks]**

Solution 3: (a)(i)

$$a_1 = \frac{1}{3^{1-1}} = 1 \qquad a_2 = \frac{1}{3^{2-1}} = \frac{1}{3} \qquad a_3 = \frac{1}{3^{3-1}} = \frac{1}{9} \qquad a_4 = \frac{1}{3^{4-1}} = \frac{1}{27}$$

$$\left\{a_1, a_2, a_3, a_4 : 1, \frac{1}{3}, \frac{1}{9}, \frac{1}{27}\right\}$$

Question 3: (a)(ii)

Show that the sequence converges. **[2 marks]**

Solution 3: (a)(ii)

$$\lim_{n\to\infty} a_n = \frac{1}{3^{n-1}}$$

$$= \frac{1}{\infty}$$

$$= 0$$

Since $\lim_{n\to\infty} a_n = l$ where $l \neq \infty$

$\implies$ The sequence converges.

Q.E.D.

Question 3: (b)(i)

Express the series $4 + 4^2 + 4^3 + \ldots + 4^n$ using summation notation. **[2 marks]**

Solution 3: (b)(i)

$$4 + 4^2 + 4^3 + \ldots + 4^n = \sum_{r=1}^{n} 4^r$$

Question 3: (b)(ii)

Prove by mathematical induction that

$$4 + 4^2 + 4^3 + \ldots + 4^n = \frac{4}{3}(4^n - 1)$$

for all positive integers n. **[8 marks]**

Solution 3: (b)(ii)

Re: $4 + 4^2 + 4^3 + \ldots + 4^n = \frac{4}{3}(4^n - 1)$

Let P_n be the statement: $\sum_{r=1}^{n} 4^r = \frac{4}{3}(4^n - 1)$ $\quad \forall n \in \mathbb{Z}^+$.

When $n = 1$

$$4^1 = \frac{4}{3}(4^1 - 1)$$

$4 = 4 \qquad$ L.H.S. = R.H.S.

$\Longrightarrow$ the statement P_1 is true

Assume true when $n = k$

$$\sum_{r=1}^{k} 4^r = \frac{4}{3}\left(4^k - 1\right)$$

R.T.P. $\sum_{r=1}^{k+1} 4^r = \frac{4}{3}\left(4^{(k+1)} - 1\right)$ when $n = k + 1$

Proof:

$$\sum_{r=1}^{k+1} 4^r = \frac{4}{3}\left(4^k - 1\right) + 4^{k+1}$$

$$\sum_{r=1}^{k+1} 4^r = \frac{4^{k+1}}{3} - \frac{4}{3} + 4^{k+1}$$

$$\sum_{r=1}^{k+1} 4^r = 4^{k+1}\left(\frac{1}{3} + 1\right) - \frac{4}{3}$$

$$\sum_{r=1}^{k+1} 4^r = 4^{k+1}\left(\frac{4}{3}\right) - \frac{4}{3}$$

$$\sum_{r=1}^{k+1} 4^r = \frac{4}{3}\left(4^{(k+1)} - 1\right)$$

Q.E.D.

But this statement is true for P_k such that if P_k is true then P_{k+1} is true. Likewise, if P_1 is true, then P_2 is true …Hence, by P.M.I. , the statement P_n is true $\forall n \in \mathbb{Z}^+$.

Question 3: (c)(i)

Determine the values of A and B such that

$$\frac{4}{(2r+1)(2r+3)} = \frac{A}{(2r+1)} + \frac{B}{(2r+3)}$$ **[3 marks]**

Solution 3: (c)(i)

$$\frac{4}{(2r+1)(2r+3)} = \frac{A}{(2r+1)} + \frac{B}{(2r+3)}$$

$$4 = A(2r+3) + B(2r+1) \qquad \times(2r+1)(2r+3)$$

When $r = -\frac{1}{2}$

$$4 = A\left[2\left(-\frac{1}{2}\right)+3\right]+0$$

$$4 = 2A$$

$$A = 2$$

When $r = 0$

$$4 = 2(0+3) + B(0+1)$$

$$4 = 6 + B$$

$$B = -2$$

$$\frac{4}{(2r+1)(2r+3)} = \frac{2}{(2r+1)} + \frac{-2}{(2r+3)}$$

Where $A = 2$ and $B = -2$

Question 3: (c)(ii)

Hence, use the method of differences to show that

$$\sum_{r=1}^{n} \frac{4}{(2r+1)(2r+3)} = 2\left(\frac{1}{3} - \frac{1}{2n+3}\right)$$ **[5 marks]**

Solution 3: (c)(ii)

Proof:

$$\text{Re:} \frac{4}{(2r+1)(2r+3)} = 2\left(\frac{1}{(2r+1)} - \frac{1}{(2r+3)}\right)$$

$$\sum_{r=1}^{n} \frac{4}{(2r+1)(2r+3)} = 2\left[\left(\frac{1}{3} - \cancel{\frac{1}{5}}\right) + \left(\cancel{\frac{1}{5}} - \cancel{\frac{1}{7}}\right) + \ldots \left(\cancel{\frac{1}{2n-1}} - \cancel{\frac{1}{2n+1}}\right) + \left(\cancel{\frac{1}{2n+1}} - \cancel{\frac{1}{2n+3}}\right)\right]$$

$$\sum_{r=1}^{n} \frac{4}{(2r+1)(2r+3)} = 2\left(\frac{1}{3} - \frac{1}{2n+3}\right) \qquad \text{By the method of differences}$$

Q.E.D.

Question 3: (c)(iii)

Hence, calculate

$$\sum_{r=1}^{\infty} \frac{4}{(2r+1)(2r+3)}$$

[3 marks]

Solution 3: (c)(iii)

Re: $\displaystyle\sum_{r=1}^{n} \frac{4}{(2r+1)(2r+3)} = 2\left(\frac{1}{3} - \frac{1}{2n+3}\right)$

$$\sum_{r=1}^{\infty} \frac{4}{(2r+1)(2r+3)} = 2\left(\frac{1}{3} - \frac{1}{(2\times\infty)+3}\right)$$

$$\sum_{r=1}^{\infty} \frac{4}{(2r+1)(2r+3)} = 2\left(\frac{1}{3} - \frac{1}{\infty}\right)$$

$$\sum_{r=1}^{\infty} \frac{4}{(2r+1)(2r+3)} = 2\left(\frac{1}{3} - 0\right)$$

$$\sum_{r=1}^{\infty} \frac{4}{(2r+1)(2r+3)} = \frac{2}{3}$$

Question 4

Question 4: (a)

Obtain the Maclaurin series expansion of $f(x) = e^{2x}$ up to and including the term in x^4. **[3 marks]**

Solution 4: (a)

Re: $f(x) = f(0) + \dfrac{f'(0)}{1!}x + \dfrac{f''(0)}{2!}x^2 + \dfrac{f'''(0)}{3!}x^3 \dots$

$f(x) = e^{2x} \qquad f(0) = 1$

$f'(x) = 2e^{2x} \qquad f'(0) = 2$

$f''(x) = 4e^{2x} \qquad f''(0) = 4$

$f'''(x) = 8e^{2x} \qquad f'''(0) = 8$

$f^{(4)}(x) = 16e^{2x} \qquad f^{(4)}(0) = 16$

$$e^{2x} = 1 + 2x + \frac{4}{2!}x^2 + \frac{8}{3!}x^3 + \frac{16}{4!}x^4 + \dots$$

$$e^{2x} = 1 + 2x + 2x^2 + \frac{4}{3}x^3 + \frac{2}{3}x^4 + \dots$$

Question 4: (b)

Obtain the binomial expansion of $(8 + x)^{\frac{1}{3}}$ in ascending powers of x, up to and including the term in x^2. State the values of x for which the expansion is valid. **[6 marks]**

Solution 4: (b)

Let: $f(x) = (8 + x)^{\frac{1}{3}}$

$$f(x) = 8^{\frac{1}{3}}\left(1 + \frac{x}{8}\right)^{\frac{1}{3}}$$

$$f(x) = 2\left[1 + \left(\frac{1}{3}\right)\frac{x}{8} + \frac{\left(\frac{1}{3}\right)\left(\frac{1}{3} - 1\right)}{2!}\left(\frac{x}{8}\right)^2 + \dots\right]$$

$$f(x) = 2\left[1 + \frac{x}{24} - \frac{x^2}{576} + \dots\right]$$

$$f(x) = 2 + \frac{x}{12} - \frac{x^2}{288} + \dots$$

The expansion is valid for: $\left|\dfrac{x}{8}\right| < 1$

$-8 < x < 8 \qquad \forall x \in \mathbb{R}$

Question 4: (c)(i)

Using the intermediate value theorem, show that $\cos x = xe^{-x}$ has a root between $x = 1$ and $x = 1.5$. **[3 marks]**

Solution 4: (c)(i)

Let: $f(x) = \cos x - xe^{-x}$

$f(1) = \cos(1) - (1)e^{-1} = 0.172$ +ve

$f(1.5) = \cos(1.5) - (1.5)e^{-1.5} = -0.264$ −ve

Since $f(1) \cdot f(1.5) < 0$, and $f(x)$ is continuous within the interval, then a root α must exist between $(1, 1.5)$.

Question (4): (c)(ii)

Hence, use the method of interval bisection to determine, correct to 1 decimal place, the approximate root of $\cos x = xe^{-x}$ which lies in the interval $(1, 1.5)$. **[6 marks]**

Solution (4): (c)(ii)

Re: $f(x) = \cos x - xe^{-x}$

$x_1 = \dfrac{1 + 1.5}{2} = 1.25$ $f(1.25) = \cos(1.25) - (1.25)e^{-1.25} = -0.0428$ −ve

Since $f(1) \cdot f(1.25) < 0$, then α must exist between $(1, 1.25)$.

$x_2 = \dfrac{1 + 1.25}{2} = 1.125$ $f(1.125) = \cos(1.125) - (1.125)e^{-1.125} = 0.066$ +ve

Since $f(1.125) \cdot f(1.25) < 0$, then α must exist between $(1.125, 1.25)$.

$x_3 = \dfrac{1.125 + 1.25}{2} = 1.1875$ $f(1.1875) = \cos(1.1875) - (1.1875)e^{-1.1875} = 0.012$ +ve

Since $f(1.1875) \cdot f(1.25) < 0$, then α must exist between $(1.1875, 1.25)$.

$x_4 = \dfrac{1.1875 + 1.25}{2} = 1.21875$ $f(1.21875) = \cos(1.21875) - (1.21875)e^{-1.21875} = -0.015$ −ve

Since $f(1.1875) \cdot f(1.21875) < 0$, then α must exist between $(1.1875, 1.21875)$.

$\alpha \approx 1.2$ (1 d.p.) By the interval bisection method

Question 4: (d)(i)

The equation $x^4 + x - 13 = 0$ has a root in the interval $(1, 2)$.

Use the Newton-Raphson formula to show that for $n \geqslant 1$, with a given initial estimate x_1,

$$x_{n+1} = \frac{3x_n^4 + 13}{4x_n^3 + 1}$$ **[4 marks]**

Solution 4: (d)(i)

Let: $f(x) = x^4 + x - 13$

$f' = 4x^3 + 1$

$$x_{n+1} = x_n - \frac{x_n^4 + x_n - 13}{4x_n^3 + 1} \qquad \text{Re: } x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

$$x_{n+1} = \frac{4x_n^4 + x_n - \left(x_n^4 + x_n - 13\right)}{4x_n^3 + 1}$$

$$x_{n+1} = \frac{3x_n^4 + 13}{4x_n^3 + 1} \qquad n \geqslant 1 \qquad n \in \mathbb{N}$$

Q.E.D.

Question 4: (d)(ii)

Hence, or otherwise, use the Newton-Raphson method with the initial estimate $x_1 = 2$ to calculate, to 2 decimal places, a new estimate, x_4, of the root. **[3 marks]**

Solution 4: (d)(ii)

$$\text{Re: } x_{n+1} = \frac{3x_n^4 + 13}{4x_n^3 + 1} \qquad x_1 = 2$$

$$x_2 = \frac{3(2)^4 + 13}{4(2)^3 + 1} = 1.8484$$

$$x_3 = \frac{3\,(1.8484)^4 + 13}{4\,(1.8484)^3 + 1} = 1.8285$$

$$x_4 = \frac{3(1.8285)^4 + 13}{4(1.8285)^3 + 1} = 1.8282$$

$x_4 = 1.83$ (2 d.p.)

Module 3

Question 5

Question 5: (a)(i)

A fair coin thrown can land on heads (H) or tails (T) and a fair die whose sides are numbered 1 to 6 are thrown simultaneously.

Determine the sample space of the possible outcomes of the experiment. [**2 marks**]

Solution 5: (a)(i)

$$S = \left\{ \begin{array}{cccccc} (H,1), & (H,2), & (H,3), & (H,4), & (H,5), & (H,6), \\ (T,1), & (T,2), & (T,3), & (T,4), & (T,5), & (T,6) \end{array} \right\} = 12 \text{ outcomes}$$

Question 5: (a)(ii)

Let F be the event that the die lands on a 5.

Calculate $\mathrm{P}(H|F)$. [**2 marks**]

Solution 5: (a)(ii)

$$\mathrm{P}(H|F) = \frac{\mathrm{P}(H \cap F)}{\mathrm{P}(F)}$$

$$\mathrm{P}(H|F) = \frac{\frac{1}{12}}{\frac{1}{6}} = \frac{1}{2}$$

$$\mathrm{P}(H \cap F) = \frac{{}^1C_1\,{}^1C_1}{12} = \frac{1}{12}$$

Question 5: (a)(iii)

State, with reason, whether (H) and (F) are independent events. [**3 marks**]

Solution 5: (a)(iii)

$$\mathrm{P}(H) \times \mathrm{P}(F) = \frac{1}{2} \times \frac{1}{6} = \frac{1}{12}$$

Re: $\mathrm{P}(H \cap F) = \frac{1}{12}$

Since $\mathrm{P}(H) \times \mathrm{P}(F) = \mathrm{P}(H \cap F) \implies$ the events (H) and (F) are independent.

Question 5: (b)(i)

Calculate the number of ways of arranging

3 letters from the first 6 letters of the alphabet. **[2 marks]**

Solution 5: (b)(i)

$^6P_3 = \dfrac{6!}{(6-3)!} = 120$ ways

$\therefore$ The number of ways of arranging 3 letters from the first 6 letters of the alphabet is 120.

Question 5: (b)(ii)

ALL the letters of the word S U C C E S S. **[3 marks]**

Solution 5: (b)(ii)

$\dfrac{7!}{3!2!} = 420$ ways

Repeated terms: CC SSS

$\therefore$ The number of ways of arranging all the letters of the word S U C C E S S is 420.

Question 5: (c)

A committee consists of 5 Mathematics teachers and 7 English teachers. Determine in how many ways a sub-committee may be formed if 2 Mathematics teachers and 3 English teachers are selected in no particular order. **[4 marks]**

Solution 5: (c)

$$\begin{array}{ccc} & \text{12 Teachers} & \\ \swarrow & & \searrow \\ \text{5 Mathematics} & & \text{7 English} \end{array}$$

No. of ways $= {}^{5}C_{2} \times {}^{7}C_{3} = \dfrac{5!}{(5-2)!2!} \times \dfrac{7!}{(7-3)!3!}$

No. of ways $= 350$

$\Longrightarrow$ 2 Mathematics and 3 English teachers can be chosen in 350 ways.

Question 5: (d)

A matrix is given by $\mathbf{A} = \begin{pmatrix} 1 & 2 & 5 \\ 3 & 0 & 2 \\ -1 & 4 & 6 \end{pmatrix}$.

Evaluate the determinant of **A**. **[4 marks]**

Solution 5: (d)

$\text{Det(A)} = 1\begin{vmatrix} 0 & 2 \\ 4 & 6 \end{vmatrix} - 2\begin{vmatrix} 3 & 2 \\ -1 & 6 \end{vmatrix} + 5\begin{vmatrix} 3 & 0 \\ -1 & 4 \end{vmatrix}$

$\text{Det(A)} = 1(0-8) - 2(18-(-2)) + 5(12-0)$

$\text{Det(A)} = -8 - 40 + 60$

$\text{Det(A)} = 12$

Question 5: (e)

A system of linear equations is given as

$$
\begin{aligned}
x + z &= 1 \\
y + z &= 2 \\
2y + z &= 1
\end{aligned}
$$

Determine whether the system is consistent. **[5 marks]**

Solution 5: (e)

Let: $A = \left(\begin{array}{ccc|c} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 2 \\ 0 & 2 & 1 & 1 \end{array}\right)$

Reducing to row echelon form:

$$A = \left(\begin{array}{ccc|c} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & -1 & -3 \end{array}\right) \qquad R_3 \to R_3 - 2R_2$$

$$A = \left(\begin{array}{ccc|c} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 1 & 3 \end{array}\right) \qquad R_3 \to -1 \times R_3$$

Since the row echelon form of the augmented matrix does not contain any contradictory rows, i.e. $\left(\begin{array}{ccc|c} 0 & 0 & 0 & k \end{array}\right)$ where $k \neq 0$, then the augmented matrix must be consistent. Thereby, the system of equations must be consistent.

Question 6

Question 6: (a)

Calculate the value of k for which the matrix $\begin{pmatrix} 1 & 5 \\ -2k & k+1 \end{pmatrix}$ is singular. **[4 marks]**

Solution 6: (a)

Let: $A = \begin{pmatrix} 1 & 5 \\ -2k & k+1 \end{pmatrix}$

$\text{Det}(A) = [1(k+1)] - [5(-2k)]$

$\text{Det}(A) = (k+1) + 10k$

$\text{Det}(A) = 11k + 1$

$11k + 1 = 0$ When matrix A is singular

$k = -\frac{1}{11}$

The matrix A is singular only when $k = -\frac{1}{11}$.

Question 6: (b)(i)

Solve the differential equation
$$\frac{\mathrm{d}y}{\mathrm{d}x} - y = x$$
to obtain the general solution. **[8 marks]**

Solution 6: (b)(i)

$$\frac{\mathrm{d}y}{\mathrm{d}x} + (-)y = x$$

is of the form $\frac{\mathrm{d}y}{\mathrm{d}x} + P(x)y = Q(x)$ Where $P(x) = -1$ $Q(x) = x$

$$\implies ye^{-x} = \int xe^{-x}\,\mathrm{d}x \qquad \text{I.F.} \rightarrow e^{\int -1\,\mathrm{d}x} = e^{-x}$$

Let:

$u = x$ $\mathrm{d}v = e^{-x}\,\mathrm{d}x$

$\frac{\mathrm{d}u}{\mathrm{d}x} = 1$ $v = -e^{-x}$

$$ye^{-x} = -xe^{-x} - \int -1e^{-x}\,\mathrm{d}x$$

$$ye^{-x} = -xe^{-x} + \int e^{-x}\,\mathrm{d}x$$

$$ye^{-x} = -xe^{-x} - e^{-x} + C \qquad C \in \mathbb{R}$$

$$y = -x - 1 + Ce^{x}$$

Question 6: (b)(ii)

Hence, given that $y = 1$ when $x = 0$, determine the particular solution. **[2 marks]**

Solution 6: (b)(ii)

Re: $y = -x - 1 + Ce^{x}$

$1 = -0 - 1 + Ce^{0}$ When $y = 1$, $x = 0$

$C = 2$

$y = -x - 1 + 2e^{x}$ Is the particular solution

Question 6: (c)(i)

A differential equation is given by $\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2} - \dfrac{\mathrm{d}y}{\mathrm{d}x} - 2y = e^{3x}$.

Obtain the complementary function of the differential equation. **[5 marks]**

Solution 6: (c)(i)

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - \frac{\mathrm{d}y}{\mathrm{d}x} - 2y = e^{3x}$$

From the AQE:

$$e^{mx}\left(m^2 - m - 2\right) = 0$$

$$m^2 - m - 2 = 0 \qquad \text{or} \qquad e^{mx} \neq 0$$

$$(m-2)(m+1) = 0$$

$$m = 2 \qquad or \qquad m = -1$$

Since $m \in \mathbb{R}$ and is distinct:

$$y_{C.F.} = Ae^{2x} + Be^{-x} \qquad A, B \in \mathbb{R}$$

Question 6: (c)(ii)

Determine a particular integral of the differential equation. **[5 marks]**

Solution 6: (c)(ii)

Let: $y = ae^{3x}$

$$\frac{\mathrm{d}y}{\mathrm{d}x} = 3ae^{3x}$$

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = 9ae^{3x}$$

$$9ae^{3x} - 3ae^{3x} - 2\left(ae^{3x}\right) = e^{3x}$$

$$9a - 3a - 2a = 1 \qquad \div\, e^{3x}$$

$$a = \frac{1}{4}$$

$$y_{P.I.} = \frac{1}{4}e^{3x}$$

Question 6: (c)(iii)

Hence, state the general solution of the differential equation. **[1 mark]**

Solution 6: (c)(iii)

$y = y_{C.F.} + y_{P.I.}$

$y = Ae^{2x} + Be^{-x} + \frac{1}{4}e^{3x}$

END